

JAVA PROGRAMMING

ASSIGNMENT 14

Problem 1: Banking System

Description:

Create a `BankAccount` class with both static and instance methods. The class should have attributes for `accountNumber`, `accountHolderName`, and `balance`. Implement static methods to track the total number of accounts and the total amount across all accounts. Implement instance methods to deposit and withdraw money, and display account details. Write a Java program to create multiple `BankAccount` objects, perform transactions, and use both static and instance methods.

Concepts Covered:

- Static vs. instance methods
- Accessing methods with objects
- Using multiple classes

Tasks:

1. Create a `BankAccount` class with attributes and methods.
2. Implement static methods to track total accounts and balance.
3. Implement instance methods for transactions and displaying details.
4. Create multiple `BankAccount` objects, perform transactions, and display results using both static and instance methods.

Problem 2: Temperature Conversion

Description:

Design a `TemperatureConverter` class with static methods for converting temperatures between Celsius and Fahrenheit. Include methods `celsiusToFahrenheit` and `fahrenheitToCelsius`. Implement a `Main` class to test these conversions and display results.

Concepts Covered:

- Static methods
- Using static methods without creating an object
- Accessing static methods in a separate class

Tasks:

1. Create a `TemperatureConverter` class with static methods for temperature conversion.
2. Implement a `Main` class to call these static methods and display conversion results.
3. Test conversions with different temperature values.

Problem 3: Rectangle Area Calculation

Description:

Create a `Rectangle` class with attributes for `length` and `width`. Implement methods to calculate the area and perimeter of the rectangle. Write a `Main` class that creates multiple `Rectangle` objects, calculates their area and perimeter, and displays the results.

Concepts Covered:

- Class methods
- Accessing methods with objects
- Using multiple classes

Tasks:

1. Create a `Rectangle` class with methods to calculate area and perimeter.
2. Create a `Main` class to instantiate `Rectangle` objects and use instance methods.
3. Display the area and perimeter for each rectangle.

Problem 4: Student Grades

Description:

Create a `Student` class with attributes for `name`, `studentId`, and an array of `grades`. Implement instance methods to calculate the average grade and determine if the student has passed (e.g., average grade ≥ 60). Implement a `Gradebook` class that manages a list of `Student` objects. Write a Java program to add students, calculate their average grades, and check if they have passed.

Concepts Covered:

- Class methods and instance methods
- Using multiple classes
- Accessing methods with objects

Tasks:

1. Create a `Student` class with methods to calculate average grade and pass/fail status.
2. Create a `Gradebook` class to manage a list of `Student` objects.
3. Write a `Main` class to add students, calculate their average grades, and display pass/fail status.

Problem 5: Company Employee Management

Description:

Design an `Employee` class with attributes for `name`, `employeeId`, and `salary`. Implement both static and instance methods. Static methods should track the total number of employees and the total payroll. Instance methods should handle salary adjustments and display employee details. Create a `Company` class to manage multiple `Employee` objects. Write a Java program to create employees, adjust salaries, and use both static and instance methods.

Concepts Covered:

- Static vs. instance methods
- Managing multiple objects and attributes
- Using multiple classes

Tasks:

1. Create an `Employee` class with static and instance methods.
2. Implement a `Company` class to manage a collection of `Employee` objects.
3. Write a `Main` class to create employees, adjust salaries, and display results using both static and instance methods.